

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Industry Problem Solving Task

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA05

COURSE OUTCOME COVERED

CO4: *Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL 5)*

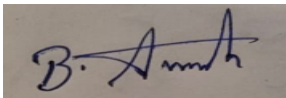
Assessment Tool Weightage:20%

Code of Conduct:

I Avinash B (192511193) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

A handwritten signature in black ink, appearing to read "B. Avinash", is written on a light-colored rectangular background.

Computer Vision & Object Detection Analysis

Questions

Comprehensive Technical Answers and Case Studies

QUESTION 21: Noise Impact on Detection

The system performance changes significantly as the noise level increases. Under low noise, the detection accuracy is 92%. At medium noise, the accuracy decreases to 80%, and at high noise it falls further to 60%. This shows a clear negative relationship between image noise and detection accuracy.

At low noise, the system performs well because important visual features remain relatively clear. The detector can distinguish objects more reliably. When the noise becomes medium, some useful features may be distorted or hidden, causing incorrect or missed detections.

At high noise, the accuracy falls to only 60%. This indicates that the system is not sufficiently robust for highly noisy environments. The reduction from 92% at low noise to 60% at high noise is 32 percentage points, which is a significant decrease.

To improve the robustness of the system, suitable preprocessing techniques such as denoising and filtering can be applied before detection. Data augmentation can also be used during training by adding different levels of noise to the training images. This allows the model to learn how to recognize objects even when the input image quality is poor.

The system should also be tested separately under low, medium, and high noise conditions. Adaptive preprocessing can be used so that stronger filtering is applied when the detected noise level is high. However, excessive filtering should be avoided because it may remove important object details.

Conclusion: *The system performs well under low noise but becomes less reliable as noise increases. Since accuracy decreases from 92% to 60%, noise reduction, noise-aware training, and suitable preprocessing should be used to improve detection robustness.*

QUESTION 24: Motion Detection under Crowd Density

The two methods behave differently as crowd density changes. Optical Flow provides high performance in sparse crowds but low performance in dense crowds. Motion Models provide moderate performance in sparse crowds and high performance in dense crowds.

In a sparse crowd, individuals and their movements are easier to distinguish. Therefore, Optical Flow can provide detailed motion information. When there are fewer people in the scene, the movement of individual objects can be estimated more easily.

However, when the crowd becomes dense, many people overlap and move close to one another. This creates complex motion patterns. As a result, Optical Flow becomes less reliable because it may not be able to distinguish individual movements clearly.

Motion Models perform better in dense crowds because they can provide more stable motion information when individual objects are difficult to separate. Although their performance is only moderate in sparse crowds, their high performance in dense crowds makes them useful for crowded environments.

For a system that needs to operate under both sparse and dense conditions, an adaptive or hybrid approach can be used. Optical Flow can be selected when the crowd is sparse, while Motion Models can be selected when the crowd becomes dense.

A crowd-density estimation module could first determine the approximate density of the scene and then automatically select the more appropriate motion technique.

Conclusion: *Motion Models are the better choice for applications mainly involving dense crowds. For a system that must work under both conditions, combining Optical Flow for sparse scenes and Motion Models for dense scenes would provide better overall performance.*

QUESTION 12: Robustness vs Speed Trade-off

The three methods have different combinations of accuracy, robustness, and speed. Method A has 88% accuracy, high robustness, and slow speed. Method B has 85% accuracy, moderate robustness, and fast speed. Method C has 90% accuracy, low robustness, and moderate speed.

Method A is highly robust, which means that its performance is expected to remain more consistent when environmental conditions change. Its accuracy of 88% is also good. However, its slow processing speed can become a limitation in real-time applications.

Method B is the fastest method and provides 85% accuracy. It is therefore useful when low latency is very important. However, its moderate robustness means that it may not perform consistently when the operating environment changes significantly.

Method C provides the highest accuracy of 90% and has moderate speed. However, it has low robustness. This means that although it can provide high accuracy under suitable conditions, its performance may decrease when the conditions become difficult.

For a balanced system, Method A is generally the safest option because high robustness is important for reliable deployment. If the application has strict real-time requirements, Method B may be preferred because of its fast speed.

Method C should be selected when accuracy is the main priority and the system can tolerate lower robustness.

Conclusion: *Method A is the best balanced choice because it provides good accuracy and high robustness. Method B should be selected when speed is the most important factor, while Method C is suitable when accuracy is prioritized over robustness.*

QUESTION 28: Motion Model Evaluation

The two motion models have different characteristics. Model A has high realism but low speed, whereas Model B has moderate realism and high speed. Therefore, the correct choice depends on the application context.

Model A is appropriate when realistic motion is the main objective. High realism is useful for applications such as cinematic animation, simulation, and high-quality offline rendering. Its low speed is less problematic in these situations because the output does not necessarily need to be generated immediately.

Model B is more suitable for real-time applications. Its high speed allows the system to generate motion quickly and maintain responsive interaction. Although its realism is moderate, the higher processing speed makes it more practical for applications where immediate response is important.

For example, a movie scene can use Model A because the rendering process can take more time to produce realistic movement. On the other hand, an interactive game needs continuous and fast motion generation, so Model B would be more suitable.

A practical system could also use both models depending on the operating mode. Model B could be used during real-time interaction, while Model A could be used for high-quality cutscenes or offline rendering.

Conclusion: *Model A is best for cinematic and offline applications where realism is the main priority. Model B is best for real-time systems where speed and responsiveness are more important.*

QUESTION 16: Object Detection under Scale Variation

The detection accuracy changes significantly depending on the size of the object. Large objects are detected with 94% accuracy, medium objects with 80% accuracy, and small objects with only 55% accuracy.

The high accuracy for large objects indicates that the system can detect objects effectively when they occupy a sufficient number of pixels in the image. More visual features are available, making detection easier.

For medium-sized objects, the accuracy decreases to 80%. This indicates that reducing object size makes feature extraction more difficult. Some important details may become less visible.

The most serious problem occurs with small objects. Their accuracy is only 55%. The difference between large and small object detection is 39 percentage points, showing that scale variation is a major limitation of the detection system.

Small objects contain fewer pixels and visual features. They can also be affected by image resolution, compression, background clutter, and occlusion.

To improve small-object detection, the system can use higher-resolution input images and multi-scale feature extraction. Image pyramids and feature-pyramid-based approaches can help the model detect objects at different scales.

Data augmentation should also include sufficient small-object examples. The training dataset should contain large, medium, and small objects under different viewing distances and environmental conditions.

The evaluation should also report detection accuracy separately for each object size rather than depending only on the overall accuracy.

Conclusion: *The system performs well for large objects but has a serious limitation for small objects. Since small-object accuracy is only 55%, improvements should focus on higher-resolution input, multi-scale processing, data augmentation, and additional small-object training samples.*

QUESTION 22: Depth Estimation Error Analysis

The three methods have different depth-estimation errors. Method A has an error of 0.5 m, Method B has an error of 1.2 m, and Method C has an error of 0.8 m. Since a lower error indicates more accurate depth estimation, Method A provides the best performance.

Method B has the highest error of 1.2 m. This means that its estimated depth can deviate more significantly from the actual distance. Such an error can be problematic in applications where accurate distance information is important.

For example, depth estimation is important in robotics, autonomous navigation, obstacle avoidance, and object measurement. In these applications, a large depth error could cause incorrect decisions.

Method C has an intermediate error of 0.8 m. It is more accurate than Method B but less accurate than Method A.

Method A has the lowest error of only 0.5 m. Therefore, it provides the most accurate depth estimation among the three methods.

For safety-critical applications where accurate distance estimation is essential, Method A should be preferred. If other practical constraints make Method A unsuitable, Method C can be considered as a compromise.

The methods should also be tested under different distances, lighting conditions, object surfaces, and camera positions before deployment. Average error alone may not completely describe the reliability of the system.

Conclusion: *Method A is the most accurate depth-estimation method because it has the lowest error of 0.5 m. Method C is the second-best option at 0.8 m, while Method B has the highest error at 1.2 m.*

QUESTION 8: Model Generalization vs Overfitting

Model A achieves 98% training accuracy but only 70% testing accuracy. Model B achieves 90% training accuracy and 88% testing accuracy. These results show a significant difference in their generalization ability.

Model A performs extremely well on the training data. However, its accuracy decreases from 98% to 70% when tested on unseen data. This is a difference of 28 percentage points.

This large difference indicates overfitting. The model may have learned patterns that are specific to the training dataset rather than learning general features that can be applied to new images.

Model B has a training accuracy of 90% and testing accuracy of 88%. The difference is only 2 percentage points. This indicates much better generalization.

Although Model B has a lower training accuracy than Model A, its testing performance is much more reliable. In a real-world application, performance on unseen data is more important than obtaining an extremely high training accuracy.

Model A can potentially be improved using regularization, data augmentation, early stopping, a larger and more diverse dataset, and appropriate validation techniques.

Data augmentation can expose the model to different viewpoints, lighting conditions, object positions, and other variations. This can help the model learn general features instead of memorizing the training examples.

Conclusion: *Model B is the more reliable model for deployment because its training and testing accuracies are very close. Model A shows strong overfitting because its accuracy falls from 98% during training to only 70% during testing.*

QUESTION 29: Detection under Occlusion

The three methods have different detection accuracies when objects are partially occluded. Method A achieves 70%, Method B achieves 85%, and Method C achieves 90%. Therefore, Method C provides the strongest performance under occlusion.

Occlusion is a challenging condition because part of an object is blocked by another object or by some element in the environment. A detection system must therefore recognize an object using only the visible portion.

Method A has the lowest accuracy at 70%, indicating that it is significantly affected by occlusion. Method B improves the accuracy to 85%, making it more reliable than Method A.

Method C achieves the highest accuracy at 90%. Therefore, it has the best performance among the three methods when objects are partially hidden.

The difference between Method A and Method C is 20 percentage points, which is a significant improvement. This indicates that Method C is more capable of extracting useful features from partially visible objects.

To further improve the system, the training dataset should contain many examples of partially occluded objects. Different levels of overlap should be included so that the model learns how objects appear when portions of them are hidden.

Suitable data augmentation techniques can also simulate occlusion during training. This can improve the model's ability to recognize objects even when important portions are not visible.

If Method C requires more computational resources, Method B can be considered as a compromise because it still provides a relatively high accuracy of 85%.

Conclusion: *Method C is the best choice for environments with significant occlusion because it provides the highest accuracy of 90%. Method B is a good alternative at 85%, while Method A is least suitable at 70%.*